

70% of workers have a college degree according to the 2017 American Community Survey (ACS).²⁵ In Figure A13 we show trends for occupations in which fewer than 30% of workers have a college degree.

Occupations with a high share of college graduates have declining employment overall, with muted differences between more-exposed and less-exposed occupations compared to our main results. In contrast, occupations with a low share of college graduates have rising overall employment, with the least AI-exposed occupations growing and the most exposed occupations declining in employment. Further, for lower college share occupations, the dispersion in employment outcomes is visible in higher age groups as well, with workers up to age 40 showing separation in employment trends by AI exposure. These findings suggest that deteriorating education outcomes cannot fully explain our main results. They also suggest that for non-college workers, experience may serve as less of a buffer to labor market disruption than for college workers.

Other Robustness Checks Figures A14 and A15 show results separately for men and women. The results are similar, suggesting that diverging prospects for men and women are not driving our findings. Figure A16 shows that results are similar when we do not take a balanced sample of firms. Figure A17 shows similar results when including part-time and temporary workers.²⁶

Comparison to CPS Data A useful benchmark for our findings is to compare them to estimates from the monthly Current Population Survey (CPS). The CPS surveys about 60,000 households nationwide each month to collect data on employment and other labor force characteristics. These data are released a few weeks after the reference month, giving close to real-time estimates of employment statistics. A number of prior analyses have used the CPS to assess how AI is impacting entry-level work (Chandar, 2025b; Dominski and Lee, 2025; Lim et al., 2025; Eckhardt and Goldschlag, 2025). We compare some of our main findings in the ADP data to estimates from the CPS.

Figures A18 through A20 show employment changes by age for software developers, customer

²⁵Not a single occupation in the first quintile of GPT-4 β based exposure measure has a college share above 35%, so that quintile is excluded from the results in Figure A12.

²⁶Another possibility is that employment trends are driven by Covid-19 Pandemic-era stimulus checks that distorted labor supply. However, these stimulus payments were conditioned on income requirements, and more AI-exposed occupations on average have higher incomes (Kochhar, 2023), suggesting the observed declines in AI-exposed occupations are unlikely to be driven by this channel.

service representatives, and home health aides by age using data from the CPS. Though there are millions of workers employed in these professions across the US, the estimates are highly volatile, with common fluctuations of 20% or greater in estimated employment month-to-month. Figure A21 shows estimated employment changes by age and exposure quintile using the CPS, also suggesting a high degree of volatility in the estimates.

This volatility in CPS microdata reflects small sample sizes and the fact that the CPS is not stratified to target employment statistics for these demographic-occupation subgroups.²⁷ The sample size and sampling procedure of the CPS may therefore make it challenging to assess employment changes by age and AI exposure with a high degree of confidence over the time horizon considered in this paper (Chandar, 2025b; O’Brien, 2025).

Other large-scale data sources such as the American Community Survey (ACS) may offer a more reliable comparison to the ADP data, though the ACS is released with a significant lag compared to the data from ADP. We encourage comparison of our findings to results from other data sources such as the ACS upon their release.²⁸

5 Conclusion

We document six facts about the recent labor market effects of artificial intelligence.

- First, we find substantial declines in employment for early-career workers in occupations most exposed to AI, such as software development and customer support.
- Second, we show that economy-wide employment continues to grow, but employment growth for young workers has been stagnant.
- Third, entry-level employment has declined in applications of AI that *automate* work, with muted effects for those that *augment* it.
- Fourth, these employment declines remain after conditioning on firm-time effects, with a 13% relative employment decline for young workers in the most exposed occupations.

²⁷The CPS includes between 26 and 53 young software developers aged between 22 and 25 per month over our sample period. It includes between 49 and 95 young customer service representatives, and between 2 and 14 young home health aides.

²⁸The 2024 ACS 1-Year Public Use Microdata Sample is scheduled to be released on October 16, 2025.

- Fifth, these labor market adjustments are more visible in employment than in compensation.
- Sixth, we find that these patterns hold in occupations unaffected by remote work and across various alternative sample constructions.

While our main estimates may be influenced by factors other than generative AI, our results are consistent with the hypothesis that generative AI has begun to significantly affect entry-level employment.

The adoption of new technologies typically leads to heterogeneous effects across workers, resulting in an adjustment period as workers reallocate from displaced forms of work to new forms with growing labor demand (Autor et al., 2024). Such endogenous adjustment may already be happening with AI, with emerging evidence of shifts in college majors away from AI-exposed categories such as computer science (Horowitz, 2025). Past transitions such as the IT revolution ultimately led to robust growth in employment and real wages following physical and human capital adjustments, with some workers benefiting more than others (Bresnahan et al., 2002; Brynjolfsson et al., 2021).

Tracking employment trends on an ongoing basis will help determine if the adjustment to AI follows a similar pattern. Consequently, we will continue to monitor these outcomes to assess whether the trends documented in the paper accelerate in the future. Future work would benefit from better firm-level AI adoption data, which would provide sharper variation for estimating plausible causal effects of AI on employment.

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